

The cost of a mark: Newton’s vanishing sagitta and a floor of order $\hbar$

Abstract. Newton reads a force off a trajectory by letting the sagitta and the area between the inertial tangent and the curve go to zero, keeping their ratio to the time. We show that the same comparison, once it must be *recorded*, has a floor. For every protocol of marks, of any number and at any times, whose resolution and recoil obey $\delta_j \Delta_j \geq \kappa$, deciding between free and forced motion over a duration τ at error probability ϵ requires

$$\tau \Delta E = \frac{F^2 \tau^3}{2m} \geq 9 z_{1-\epsilon}^2 \kappa,$$

and for a mark realized by coupling the body’s position to a probe, the error operator and the impulse delivered are canonically conjugate, so $\kappa \geq \hbar/2$ exactly. The floor is about $12\hbar$ at five per cent error, and insertion of marks finer than $\tau_* = (9z_{1-\epsilon}^2 m \hbar / F^2)^{1/3}$ records free motion only. The historical claim is that Newton’s *Opticks* contains both factors of κ , one of them measured as the 1/89000th part of an inch, and that the single proposition joining them to a floor is an indeterminacy about the fits which Newton formulated the negation of. The distance between his system and a positive $\hbar$ is therefore one nameable proposition, and that proposition is Robertson’s inequality for his corpuscle.

Draft, 2026-09-17. Synthesis of the mark-floor, derivation, probabilistic and mark-cost notes, which hold the proofs in full. Two obligations remain before submission and are stated in §8. Exploratory; no ledger promotion.

1. The question

Galileo’s comparison is an inertial horizontal line against a falling parabola. With mass m , transverse force F , horizontal speed v and duration τ , the two paths are $q_I(t) = (vt, 0)$ and $q_F(t) = (vt, Ft^2/2m)$, and the region between them has

$$A = \frac{vF\tau^3}{6m}, \quad s(\tau) = \frac{F\tau^2}{2m}, \quad \Delta E = \frac{F^2\tau^2}{2m}, \quad \frac{3F}{v}A = \tau\Delta E. \quad (1)$$

The sagitta s , the area A and the product $\tau\Delta E$ all scale as powers of τ and vanish together. Newton’s Lemma X makes the first displacement quadratic in the time; Lemma XI, Corollaries 4 and 5, make the curved segment one third of the tangent triangle with both cubic in the time; Proposition VI reads the force off the limit $2ms/\tau^2$; and Proposition I builds an orbit as a polygon struck by impulses at its vertices, the number of vertices “augmented in infinitum”. The construction is designed so that the quantities in (1) disappear and their ratios survive.

Our question is what stops that refinement when the trajectory has to be recorded rather than contemplated, and whether the stopping point can be reached from materials Newton possessed.

The answer has three parts. The geometry alone supplies no floor (§2). Recording supplies one, which is set by a single number κ with the dimensions of action, and every protocol of marks faces the same bound in the same combination $F^2\tau^3/m$ (§3). Quantum kinematics fixes $\kappa \geq \hbar/2$ by an exact and elementary argument (§4). Newton’s optics supplies both factors of κ and denies the inequality between them (§§5–6).

2. The geometry has no floor

Proposition 1. For every partition $0 = t_0 < \dots < t_n = \tau$ with steps τ_j and mesh $|\pi|$, the vertex impulse $F\tau_j$, the step sagitta $F\tau_j^2/2m$ and the step area $vF\tau_j^3/6m$ tend to zero with the mesh; the areas sum to at most $vF\tau|\pi|^2/6m$; and $2m s(\tau_j)/\tau_j^2 = F$ exactly at every step.

The proof is immediate from (1). Berkeley's objection, that evanescent increments are "neither finite Quantities nor Quantities infinitely small, nor yet nothing", is answered inside the geometry by the limit ratio, which is what Newton's closing Scholium to Book I Section I provides. A floor must therefore come from the act of marking the curve, and the reading of "inserting a point" used here is physical: a point of the trajectory is inserted by making a mark on the body at a time.

3. Marks, and the floor they impose

Definitions. A **mark** at time t_j leaves a datum about the transverse coordinate. Its **resolution** δ_j is the spread of the resulting position estimate's error; its **recoil** Δ_j is the spread of the transverse impulse it delivers to the body. A protocol is a finite set of marks; it **decides** the comparison at error probability ϵ when some test on the full record does so with equal priors. A model satisfies the **mark trade-off** with constant κ when every available mark has $\delta\Delta \geq \kappa$.

The statistical model. Mark j returns $R_j = y(t_j) + \xi_j$ with ξ_j centred Gaussian of standard deviation δ_j , and delivers an impulse ι_j , centred Gaussian of standard deviation Δ_j , independently, so that

$$y(t) = y_0 + v_0 t + \theta P(t) + \frac{1}{m} \sum_j \iota_j (t - t_j)_+, \quad P(t) = \frac{Ft^2}{2m},$$

with $\theta = 0$ under the inertial hypothesis and $\theta = 1$ under the falling one, and y_0, v_0 unknown. Tests are invariant under the two unknowns, so the statistic is $u^\top R$ with $\sum_i u_i = 0$ and $\sum_i u_i t_i = 0$, and the optimal invariant test has error probability $\Phi(-d/2)$ with $d^2 = \sup(u^\top P)^2 / (u^\top \Sigma u)$ over such u .

Theorem 2. For every protocol satisfying the mark trade-off with constant κ ,

$$d^2 \leq \frac{2F^2\tau^3}{9m\kappa},$$

independently of the number of marks and of their times. Hence deciding at error probability ϵ requires

$$F^2\tau^3 \geq 18z_{1-\epsilon}^2 m\kappa, \quad \tau\Delta E \geq 9z_{1-\epsilon}^2 \kappa, \quad A \geq \frac{6z_{1-\epsilon}^2 v\kappa}{F},$$

with $z_{1-\epsilon} = \Phi^{-1}(1 - \epsilon)$.

Proof. The noise enters R in two ways: ξ_j affects R_j alone, and ι_j affects R_i for $i > j$ with coefficient $(t_i - t_j)/m$. Hence

$$u^\top \Sigma u = \sum_j \delta_j^2 u_j^2 + \sum_j \frac{\Delta_j^2}{m^2} S_j^2, \quad S_j = \sum_{i>j} u_i (t_i - t_j),$$

and the arithmetic-geometric mean inequality applied termwise gives $u^\top \Sigma u \geq \frac{2\kappa}{m} \sum_j |u_j| |S_j|$.

Introduce $N(w) = \sum_{i:t_i > w} u_i$ and $T(w) = \int_w^\tau N$. Since $\sum_i u_i = 0$, N vanishes for $w < 0$ and for $w \geq t_k$; T is continuous, piecewise linear, constant on $(-\infty, 0]$, zero on $[\tau, \infty)$, with $T' = -N$ jumping by u_j at t_j and $T(t_j) = S_j$. Two identities follow. Fubini and one integration by parts, using $P'(0) = 0$ and $T(\tau) = 0$, give

$$u^\top P = \int_0^\tau P' N = - \int_0^\tau P' T' = \int_0^\tau P'' T = \frac{F}{m} \int_0^\tau T;$$

and since T' has bounded variation with jumps u_j and no other variation, while TT' vanishes at both ends,

$$\sum_j u_j T(t_j) = \int_{\mathbb{R}} T dT' = - \int_{\mathbb{R}} T'^2 = -E, \quad E = \int_0^\tau T'^2.$$

Therefore $\sum_j |u_j| |S_j| \geq E$ and $u^\top \Sigma u \geq 2\kappa E/m$. Cauchy–Schwarz with $T(\tau) = 0$ gives $|T(w)| \leq \sqrt{\tau - w} \sqrt{E}$, hence $\int_0^\tau T \leq \frac{2}{3} \tau^{3/2} \sqrt{E}$, and

$$\frac{(u^\top P)^2}{u^\top \Sigma u} \leq \frac{(F/m)^{2/3} \tau^3 E}{2\kappa E/m} = \frac{2F^2 \tau^3}{9m\kappa}.$$

Finally $\Phi(-d/2) \leq \epsilon$ requires $d \geq 2z_{1-\epsilon}$. $\square$

Three features of the proof carry the paper’s claim to universality. The certificate u ranges over all invariant tests, so no monitoring scheme is privileged. The function T encodes the whole protocol in one object, and the bound depends on it only through the Dirichlet energy E , which cancels. And the exponent 3 on τ comes from a single Cauchy–Schwarz inequality against $\sqrt{\tau - w}$, which is where the combination in (1) originates.

Corollary 3 (insertion). Marks inside a window of duration τ' , used by themselves, record the force only if $F^2 \tau'^3 \geq 18z_{1-\epsilon}^2 m\kappa$. So insertion stops at

$$\tau_* = \left(\frac{18z_{1-\epsilon}^2 m\kappa}{F^2} \right)^{1/3},$$

and finer marks are consistent with free motion at the stated confidence. Newton’s polygon may be refined past τ/τ_* vertices, and its impulses remain in the geometry, though no record exhibits them.

A worst-case counterpart. If the marks instead confine position and impulse to intervals with certainty, the same three steps run with a separating hyperplane in place of the optimal test direction and give $F^2 \tau^3 > 9m\kappa$ for every protocol, with two marks sufficing above $36m\kappa$. That version has no quantum instance, since certain position confinement forces full-support momentum by Paley–Wiener; it is recorded here because it shows that the argument depends on the quadratic structure rather than on any probabilistic assumption.

4. The cost of a mark is the probe’s uncertainty product

Theorem 4. Let a mark be realized by the impulsive coupling of the body’s position to a probe, $U = \exp(-\frac{i}{\hbar} \lambda \hat{y} \hat{P}_A)$ with $[\hat{Q}_A, \hat{P}_A] = i\hbar$, the pointer $\hat{Q}_A$ being read afterwards. Then

$$U^\dagger \hat{Q}_A U = \hat{Q}_A + \lambda \hat{y}, \quad U^\dagger \hat{p} U = \hat{p} - \lambda \hat{P}_A,$$

so the error operator $\hat{N} = \hat{Q}_A/\lambda$ and the delivered impulse $\hat{D} = -\lambda \hat{P}_A$ satisfy $[\hat{N}, \hat{D}] = -i\hbar$ and

$$\kappa = \delta \Delta = \Delta \hat{N} \cdot \Delta \hat{D} = \Delta \hat{Q}_A \cdot \Delta \hat{P}_A \geq \frac{\hbar}{2}$$

for every probe state, pure or mixed.

Proof. With $A = \frac{i}{\hbar} \lambda \hat{y} \hat{P}_A$ the commutators $[A, \hat{Q}_A] = \lambda \hat{y}$ and $[A, \hat{p}] = -\lambda \hat{P}_A$ are central, so the Baker–Campbell–Hausdorff series terminates and gives the two displayed relations, while $\hat{y}$ and $\hat{P}_A$ are unchanged. Estimating the body’s position by $\hat{Q}_A/\lambda$ leaves the error $\hat{N} = \hat{Q}_A/\lambda$, and $[\hat{N}, \hat{D}] = -[\hat{Q}_A, \hat{P}_A] = -i\hbar$. Robertson’s inequality closes it. $\square$

The coupling strength cancels, so the cost is a property of the mark rather than of how hard it is made. The record and the recoil are themselves incompatible, since $[\hat{Q}_A + \lambda \hat{y}, -\lambda \hat{P}_A] = -i\hbar \lambda$, which is why no reading of the pointer determines the impulse. A deterministic probe, whose later position records fix its momentum, has $\kappa = 0$ and no floor; this is the exact point at which classical record models differ.

For Gaussian probe states and the linear dynamics of the comparison, the Wigner function is a probability density evolving classically, so the statistical model of §3 reproduces the quantum experiment exactly. With Theorem 4,

$$\tau \Delta E \geq \frac{9}{2} z_{1-\epsilon}^2 \hbar, \quad \tau_* = \left(\frac{9 z_{1-\epsilon}^2 m \hbar}{F^2} \right)^{1/3}, \quad (2)$$

about $12\hbar$ at $\epsilon = 0.05$ and $4.5\hbar$ at $\epsilon = 0.32$.

5. Both factors are in the *Opticks*

Newton’s optics supplies the two spreads whose product is κ , in the form of two least quantities of the probe.

A least length, measured. Book II Part III defines the interval of the fits as “the space it passes between every return and the next return” of the ray’s disposition to be reflected, and Proposition XVIII gives it: for the confine of yellow and orange passing perpendicularly into air, “the Intervals of their Fits of easy Reflexion are the 1/89000th part of an Inch”, that is about $0.285 \mu\text{m}$. A datum obtained through the fits repeats with that period, so it locates the corpuscle no better; Newton’s rings are a gauge of exactly this kind. This is $\Delta \hat{Q}_A$.

A least impulse, posited. Query 29 asks whether the rays of light are “very small Bodies emitted from shining Substances”, which gives each a transverse momentum scale. Newton has no way to measure it. This is $\Delta \hat{P}_A$.

Colour dependence, measured. Observation 13 of Book II Part I reports the ratio of the red to the violet interval as “greater than as 3 to 2, and less than as 13 to 8”, and “By the most of my Observations it was as 14 to 9”. Query 29 makes the violet corpuscles the least and the red the biggest. The product $\Delta \hat{Q}_A \Delta \hat{P}_A$ is therefore colour dependent on Newton-age evidence, and its universality across the spectrum is what Planck’s constant later supplies.

Inflexion. Query 1 asks whether bodies “act upon Light at a distance, and by their action bend its Rays”, most strongly at the least distance, which is the effect that limits a mark’s resolution when the beam is narrowed to improve it.

All five passages are held verbatim with line anchors in the source companion, from the fourth edition of 1730.

6. The junction, and the proposition Newton denied

Theorem 4 says what the missing premise is. In the mark-floor note it was stated as M3, that a mark fixing the position within δ leaves the delivered impulse undetermined within κ/δ . Theorem 4 identifies κ as the probe’s own uncertainty product, so

$$\text{M3 for Newton’s corpuscle} \iff \Delta \hat{Q}_A \cdot \Delta \hat{P}_A \geq \text{a positive constant},$$

which is Robertson’s inequality stated in Newton’s two quantities. With M3, §§3–4 give the floor and the insertion mesh from Newton’s own materials, with $\kappa = \Lambda p$ in place of $\hbar/2$; the identification $\Lambda = \lambda/2$, $p = h/\lambda$ makes $\Lambda p = h/2$, which differs from $\hbar/2$ by π .

Newton denied M3, and the denial is explicit rather than inferred. Proposition XII of Book II Part III states that every ray is “put into a certain transient Constitution or State, which in the progress of the Ray returns at equal Intervals, and disposes the Ray at every return to be easily transmitted”. The disposition is a determinate periodic property carried by the ray, so its fate at a surface is fixed by its phase, and what a prior record fails to supply is knowledge rather than determination. On that reading a probe’s position and momentum are both definite, later records can recover the impulse it delivered, $\kappa = 0$, and the comparison has no floor. The countermodel is explicit: let the corpuscle travel freely after the mark to a screen at distance

D , and two position records fix its transverse momentum to within $p(\delta + \delta_s)/D$, which vanishes as D grows.

The historical claim is therefore narrow and checkable. Newton’s optics contains both factors of the product that bounds the recorded comparison; his mathematics contains the comparison and takes it to zero; and the one proposition that would join them is an indeterminacy about the fits whose negation he asserted in print. The gap between the *Principia*’s vanishing sagitta and the *Opticks*’ finite interval of fits follows from that single commitment, and the separation of the two books is incidental to it. No counterfactual about what Newton might have discovered is needed, and none is offered.

7. What the theorem does not give

Preparations of unbounded extent. Theorem 2 concerns protocols that mark the trajectory. A protocol that prepares once, waits and measures once faces a different bound. In that case the two hypotheses differ by the phase-space displacement $(-Ft^2/2m, Ft)$, whose components have product exactly $\tau\Delta E$, and Mandelstam–Tamm with Helstrom gives

$$\frac{F\tau L}{\hbar} + \frac{F\tau^2 P}{2m\hbar} \geq 1 - 2\epsilon, \quad \tau\Delta E \geq \frac{(1 - 2\epsilon)^2 \hbar^2}{4A},$$

for an apparatus whose position and momentum spreads are bounded by L and P with $A = LP \geq \hbar/2$; and this holds for every finite adaptive protocol of instruments, because the displacement is a phase-space path whose total variation does not grow when it is subdivided. The bound is tight: two narrow packets separated by $\pi\hbar/(F\tau)$ decide the comparison with error probability of order $\tau\Delta E/\hbar$, for arbitrarily small $\tau\Delta E$, at the price of an apparatus whose extent diverges. So the floor of order $\hbar$ requires either that the trajectory be marked or that the laboratory be bounded, and a non-Gaussian preparation of unbounded extent evades it.

Prior art. The combination $F^2\tau^3 \gtrsim m\hbar$ is the standard quantum limit for detecting a force on a free mass (Braginsky and Khalili, *Quantum Measurement*, 1992), and its status has been disputed since Yuen’s objection: Caves defended it (PRL **54**, 2465, 1985) and Ozawa exhibited a measurement breaking it for free-mass position (PRL **60**, 385, 1988). The contribution here is universality over protocols with an explicit constant, the exact mark cost of Theorem 4, and the identification of the premise. The inequality itself is not new.

Instruments outside the momentum-transfer class. Theorem 4 holds where $[\hat{N}, \hat{D}] = -i\hbar$. Ozawa’s counterexamples to the naive Heisenberg product concern instruments where that commutator differs, and a general class would need the proved calibration relation of Busch, Lahti and Werner (PRL **111**, 160405, 2013) or an argument of another kind. No error–disturbance relation is used above.

8. Obligations before submission

Two, both on the history side, and both stated so that a reader can see what the present draft rests on.

1. **Historiography, to be read rather than cited.** Shapiro’s *Fits, Passions, and Paroxysms* (1993) on the theory of fits and how determinate Newton meant it, which is where §6’s central claim must be tested; Sabra’s *Theories of Light from Descartes to Newton* (1981) for the optical context of Query 29; Guicciardini’s *Reading the Principia* (1999) for what the limit arguments of §2 were for.
2. **Editions.** The *Opticks* passages come from a transcription of the 1730 fourth edition and should be cited from that printing; the *Principia* passages should be cited from Cohen and Whitman in place of the Motte text used here.

Three mathematical items remain open and are not obligations of the paper: extending Theorem 4 beyond the momentum-transfer class; the general force law, for which the proof of Theorem 2 already pairs $\int_0^\tau P''T$ against the Dirichlet energy and so bounds a norm of P'' ; and closing the constants between the worst-case form, the statistical form of Theorem 2 and the single-shot form of §7.

9. Consequence for STATE

This is the synthesis the goal asked for: the Planck gap established with Newton-age arguments and their modern equivalents, in one document whose historical and technical halves are load-bearing for each other. The foundations content is Theorems 2 and 4 with §7's honest positioning; the history content is §5 and §6, resting on the source companion; the junction is the identification of M3 with Robertson's inequality for the corpuscle. STATE's queue reduces to §8's two obligations and the three open mathematical items.